

Calculations involving equilibrium reactions (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to solve problems involving values of the equilibrium constant, K_c , and initial and equilibrium concentrations of the components of an equilibrium mixture.

Have you ever tried walking up the down escalator in your local shopping centre? Let's hope not because not only is it against the rules but it is also very dangerous! But let's imagine you did try it. What would happen if you walked up at the same speed as the escalator is coming down? How would you get to the top?

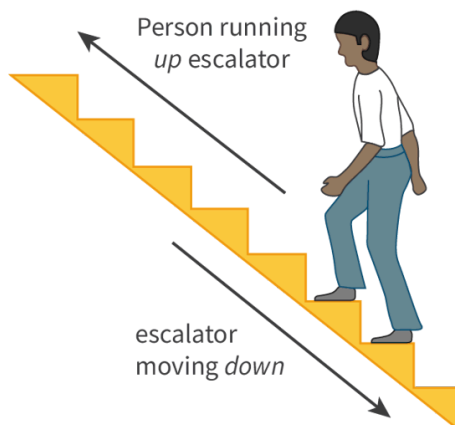


Figure 1. How is walking up an escalator that is moving down analogous to a reaction at equilibrium?

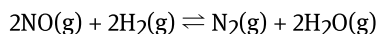
If you walked up the escalator at the same speed as the escalator was coming down, then you would stay in the same position, which is analogous to a chemical reaction at equilibrium. To get to the top, you would have to run at a faster speed than the escalator is coming down. This is analogous to the forward reaction happening faster than the reverse reaction.

In section R2.3.2–3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-equilibrium-constant-kc-and-magnitude-of-the-kc-id-46067/>) you learned that the equilibrium law allows the equilibrium constant, K_c , to be determined from the stoichiometry of the balanced equation for a reaction. In this section, you will learn how to calculate equilibrium concentrations of reactants and products of an equilibrium mixture.

Calculating equilibrium concentrations from initial concentrations

In the first example, you are given the initial concentrations of the reactants and products and the equilibrium concentration of at least one product. From this data, the equilibrium concentrations of the other species can be calculated, together with the equilibrium constant, K_c , for the reaction.

Nitrogen monoxide reacts with hydrogen to produce nitrogen and water according to the following equation:



0.100 mol of NO, 0.0500 mol of H_2 and 0.100 mol of H_2O are placed in a sealed 1.00 dm^3 flask at 25°C and allowed to reach equilibrium.

At equilibrium, 0.0620 mol of NO is present in the flask.

Determine the equilibrium concentrations of H_2 , N_2 and H_2O and calculate the value of the equilibrium constant, K_c .

Study skills

In these examples, you are given the amount (in mol) of reactants or products and the volume of the reaction vessel (usually 1.00 dm^3). To calculate the concentrations [section S1.4.5](#) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-concentration-of-a-solution-id-44263/>) of reactants and products, simply divide the amount (in mol) by the volume of the reaction vessel (in dm^3). Always check the volume of the reaction vessel in the question and do not assume that it is always going to be 1.00 dm^3 .

To calculate the equilibrium concentration of reactants and products we can make an ICE box. ICE stands for the initial concentration, change in concentration and equilibrium concentration, all in mol dm^{-3} . Once the ICE box is made, input the known concentrations. Note that if no value is given in the question, we assume the initial concentration is zero.

	2NO(g)	2H ₂ (g)	⇌
Initial concentration (mol dm^{-3})	0.100	0.0500	
Change in concentration (mol dm^{-3})			
Equilibrium concentration (mol dm^{-3})	0.0620		

Using the coefficients in the balanced equation, we can deduce that the change in the concentrations of H_2 and H_2O must be numerically the same as the change for the NO (0.0620 mol dm^{-3}). This is because the coefficients of the H_2 and H_2O are the same. The change in the concentration of the N_2 will be half this value because the coefficient is a one (which is half of two). Note, however, that the sign of the change

will be the opposite; the sign for the reactants will be negative because they are decreasing and the sign for the products are positive because they are increasing. The completed ICE box is shown below.

	2NO(g)	2H ₂ (g)	⇌
Initial concentration (mol dm ⁻³)	0.100	0.0500	
Change in concentration (mol dm ⁻³)	-0.0380	-0.0380	
Equilibrium concentration (mol dm ⁻³)	0.0620	0.0120	

Now that we have the equilibrium concentrations of the reactants and products we can calculate the value of the equilibrium constant.

First, write the equilibrium constant equation for the reaction:

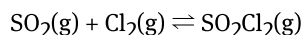
$$K_c = \frac{[\text{N}_2][\text{H}_2\text{O}]^2}{[\text{NO}]^2[\text{H}_2]^2}$$

Next, substitute in the concentrations of reactants and products and calculate the value of K_c .

$$\begin{aligned} K_c &= \frac{[0.0190][0.138]^2}{[0.0620]^2[0.0120]^2} \\ &= 653.681\dots \\ &= 654 \text{ (3 s.f.)} \end{aligned}$$

Worked example 1

0.80 mol of SO₂(g) and 1.60 mol of Cl₂(g) are placed in a 1.00 dm³ sealed flask and allowed to reach equilibrium.



At equilibrium 0.40 mol of SO₂Cl₂(g) are present.

Determine the equilibrium concentrations of SO₂(g) and Cl₂(g) and calculate the value of the equilibrium constant, K_c .

Construct an ICE box and determine the equilibrium concentrations of SO₂(g) and Cl₂(g).

	SO ₂ (g)	Cl ₂ (g)
Initial concentration (mol dm ⁻³)	0.80	1.60
Change in concentration (mol dm ⁻³)	-0.40	-0.40

	SO₂(g)	
Equilibrium concentration (mol dm ⁻³)	0.40	

Write the equilibrium constant expression for the reaction:

$$K_c = \frac{[\text{SO}_2\text{Cl}_2]}{[\text{SO}_2][\text{Cl}_2]}$$

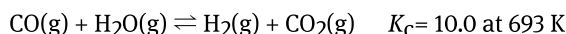
Next, substitute in the concentrations of reactants and products and calculate the value of K_c .

$$\begin{aligned} K_c &= \frac{[0.40]}{[0.40][1.20]} \\ &= 0.833\dots \\ &= 0.83 \text{ (2 s.f.)} \end{aligned}$$

Calculating equilibrium concentrations from initial concentrations and the equilibrium constant, K_c , for the reaction

In the next example, we will calculate equilibrium concentrations of reactants and products from initial concentrations and the equilibrium constant, K_c , for the reaction. The equilibrium concentrations of reactants and products can be determined by constructing an ICE box and solving for the unknown value, x .

Carbon monoxide reacts with steam according to the following equation:



1.00 mol of CO(g) and 1.00 mol of H₂O(g) are mixed in a 1.00 dm³ container. Calculate the concentration of each component in the equilibrium mixture.

First, construct an ICE box and input the initial concentrations of CO and H₂O. Because we do not know the equilibrium concentrations of any of the reactants or products, we will use x to represent the change. The change in concentration for CO(g) and H₂O(g) are both $1.00 - x$ and the change in concentration for H₂(g) and CO₂(g) are both $+x$.

	CO(g)	H₂O(g)	
Initial concentration (mol dm ⁻³)	1.00	1.00	
Change in concentration (mol dm ⁻³)	-x	-x	
Equilibrium concentration (mol dm ⁻³)	1.00 - x	1.00 - x	

Next, substitute these values into the expression for the equilibrium constant and solve for x .

$$K_c = \frac{[\text{H}_2][\text{CO}_2]}{[\text{CO}][\text{H}_2\text{O}]}$$

$$= \frac{[x][x]}{[1.00 - x][1.00 - x]}$$

Substitute in the value of the equilibrium constant:

$$10.0 = \frac{x^2}{(1.00 - x)^2}$$

This is a quadratic equation, however there is a way to simplify this equation by taking the square root of both sides of the equation.

Take the square root of both sides of the equation:

$$\sqrt{10.0} = \sqrt{\frac{x^2}{(1.00 - x)^2}}$$

Solve for x :

$$\frac{x}{(1.0 - x)} = 3.16 \dots$$

$$x = 0.760$$

Finally, we use the value of x to determine the equilibrium concentrations of reactants and products.

$$[\text{CO}] = 1.00 - 0.760 = 0.240 \text{ mol dm}^{-3}$$

$$[\text{H}_2\text{O}] = 1.00 - 0.760 = 0.240 \text{ mol dm}^{-3}$$

$$[\text{H}_2] = x = 0.760 \text{ mol dm}^{-3}$$

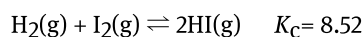
$$[\text{CO}_2] = x = 0.760 \text{ mol dm}^{-3}$$

Study skills

Note that the use of quadratic equations will not be assessed in IB DP chemistry.

Worked example 2

Calculate the equilibrium concentrations of $\text{H}_2(\text{g})$ and $\text{I}_2(\text{g})$ for this reaction if the initial concentrations are both 0.15 mol dm^{-3} .



Construct an ICE box for the reaction.

	$\text{H}_2(\text{g})$	I_2
Initial concentration (mol dm^{-3})	0.15	C
Change in concentration (mol dm^{-3})	$-x$	.

	H₂(g)	I₂
Equilibrium concentration (mol dm ⁻³)	0.15-x	0.1

Next, write the equilibrium constant expression:

$$\begin{aligned}
 K_c &= \frac{[\text{HI}]^2}{[\text{H}_2][\text{I}_2]} \\
 &= \frac{[2x][2x]}{[0.15-x][0.15-x]} \\
 8.52 &= \frac{4x^2}{(0.15-x)^2}
 \end{aligned}$$

This is a quadratic equation, however there is a way to simplify this equation by taking the square root of both sides of the equation.:

$$\begin{aligned}
 \sqrt{8.52} &= \sqrt{\frac{4x^2}{(0.15-x)^2}} \\
 \frac{2x}{0.15-x} &= 2.92 \\
 x &= 0.089
 \end{aligned}$$

$$[\text{H}_2] = 0.15 - 0.089 = 0.061 \text{ mol dm}^{-3}$$

$$[\text{I}_2] = 0.15 - 0.089 = 0.061 \text{ mol dm}^{-3}$$

$$[\text{HI}] = 2(0.089) = 0.18 \text{ mol dm}^{-3}$$

Calculating equilibrium concentrations when the value of the equilibrium constant, K_c , is very small

In the final example we will calculate the equilibrium concentrations of reactants and products where the value of the *equilibrium constant* is very small. The small value means that the forward reaction barely proceeds so the concentration of products is very small. For this type of reaction, the approximation used is $[\text{reactant}]_{\text{initial}} \approx [\text{reactant}]_{\text{eqm}}$. This means that the equilibrium concentration of the reactants is approximately equal to the initial concentration.

Nature of Science

Aspect: Measurement

How valid are assumptions made in science? For example, the mass of an electron is often assumed to be negligible compared to the mass of a proton or a neutron. But how valid is this assumption? The mass of an electron is 9.109384×10^{-31} kg and the mass of a proton is 1.672622×10^{-27} kg. So a single proton has the same mass as approximately 1836 electrons. In this example, we can say the assumption is valid, which is why it is said that almost all of the mass of an atom is contained in its dense nucleus. Can you think of any other assumptions made in science? How valid are they?

Bromine and chlorine react to form bromine chloride as follows:



0.80 mol of $\text{Br}_2(\text{g})$ and 0.80 mol of $\text{Cl}_2(\text{g})$ are added to a 1.00 dm^3 reaction vessel and allowed to reach equilibrium. Calculate the equilibrium concentration of $\text{BrCl}(\text{g})$. Because the value of the K_c is very small, the assumption is $[\text{reactants}]_{\text{initial}} \approx [\text{reactants}]_{\text{eqm}}$. So the equilibrium concentrations of $\text{Br}_2(\text{g})$ and $\text{Cl}_2(\text{g})$ are assumed to be 0.80 mol dm^{-3} .

Start by constructing an ICE box and input the known values.

	$\text{Br}_2(\text{g})$	$\text{Cl}_2(\text{g})$	$\rightleftharpoons$	$2\text{BrCl}(\text{g})$
Initial concentration (mol dm^{-3})	0.80	0.80		
Change in concentration (mol dm^{-3})	$0.80 - x$	$0.80 - x$		+
Equilibrium concentration (mol dm^{-3})	≈ 0.80	≈ 0.80		;

Write the equilibrium constant equation:

$$K_c = \frac{[\text{BrCl}]^2}{[\text{Br}_2][\text{Cl}_2]}$$

Substitute in the known values and solve for x:

$$\begin{aligned} 8.00 \times 10^{-9} &= \frac{[2x]^2}{[0.80][0.80]} \\ 4x^2 &= 5.12 \times 10^{-9} \\ x &= 3.577 \times 10^{-5} \\ &= 3.6 \times 10^{-5} \\ &\text{(2s. f.)} \end{aligned}$$

The equilibrium concentration of $\text{BrCl}(\text{g})$ is $2x$ so:

$$[\text{BrCl}(\text{g})] = 2 \times 3.6 \times 10^{-5} = 7.2 \times 10^{-5} \text{ mol dm}^{-3}$$

The very low concentration of the product is logical because the equilibrium constant value was very small, meaning that the equilibrium position lies very much to the left.

If we make the ICE box again, but this time subtract x from the initial concentrations, we can see that the equilibrium concentration of Br_2 and Cl_2 are both 0.80 mol dm^{-3} when rounded to two significant figures. Therefore, the assumption that was made in the earlier calculation was valid.

	$\text{Br}_2(\text{g})$	$\text{Cl}_2(\text{g})$	$\rightleftharpoons$	$2\text{BrCl}(\text{g})$
Initial concentration (mol dm^{-3})	0.80	0.80		C

	$\text{Br}_2(\text{g})$	$\text{Cl}_2(\text{g})$	$\rightleftharpoons$	$2\text{BrCl}(\text{g})$
Change in concentration (mol dm^{-3})	$0.80 - 3.6 \times 10^{-5}$	$0.80 - 3.6 \times 10^{-5}$		$2 \times 3.6 \times 10^{-5}$
Equilibrium concentration (mol dm^{-3})	0.80	0.80		7.2×10^{-5}

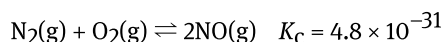
Study skills

The assumption $[\text{reactant}]_{\text{initial}} \approx [\text{reactant}]_{\text{eqm}}$ is used when the value of a very small number is added or subtracted from a larger value and there is no change in the value of the larger number when rounded to the appropriate number of significant figures. For example, when calculating the pH of a weak acid or base in [subtopic R3.1](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/>).

Worked example 3

The reaction shown between nitrogen and oxygen is shown:



The initial concentrations of N_2 and O_2 are $0.033 \text{ mol dm}^{-3}$ and $0.0081 \text{ mol dm}^{-3}$ respectively.

Determine the equilibrium concentration of $\text{NO}(\text{g})$ and include any assumptions made in your calculations.

Construct an ICE box and input the initial concentrations of N_2 and O_2 .

Because the equilibrium constant is very small the assumption is $[\text{reactant}]_{\text{initial}} \approx [\text{reactant}]_{\text{eqm}}$.

	$\text{N}_2(\text{g})$	O_2
Initial concentration (mol dm^{-3})	0.033	0.0081
Change in concentration (mol dm^{-3})	$0.033 - x$	$0.0081 - x$
Equilibrium concentration (mol dm^{-3})	≈ 0.033	≈ 0.0081

Write the equilibrium constant equation for the reaction.

$$K_c = \frac{[\text{NO}]^2}{[\text{N}_2][\text{O}_2]}$$

Substitute in the values and solve for x .

$$4.8 \times 10^{-31} = \frac{[2x]^2}{[0.033][0.0081]}$$

$$4x^2 = 1.28 \times 10^{-34}$$

$$x = 5.7 \times 10^{-18} \text{ (2 s.f.)}$$

$$[\text{NO(g)}] = 2x$$

$$= 2 \times 5.7 \times 10^{-18}$$

$$= 1.1 \times 10^{-17} \text{ mol dm}^{-3}$$

The assumption made in this calculation is that the equilibrium concentration is the same as the initial concentration.

The value of the equilibrium constant is very small meaning that the forward reaction has barely proceeded.

Making connections

How does the equilibrium law help us to determine the pH of a weak acid, weak base or a buffer solution?

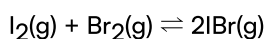
When calculating the pH of a weak acid, the assumption is that the equilibrium concentration of the weak acid is equal to the initial concentration of the weak acid. [Subtopic R3.1](#)

(<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43484/>)

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 25 minutes
- **Activity type:** Individual activity

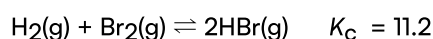
1. Iodine and bromine react to form iodine bromide according to the following equation.



0.40 mol of I_2 and 0.40 of Br_2 are mixed in a 1.00 dm^3 vessel. At equilibrium, 0.40 mol of IBr are present.

Calculate the value of the equilibrium constant for the reaction.

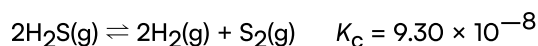
2. Hydrogen and bromine react to form hydrogen bromide according to the following equation.



0.600 mol of H_2 and 0.600 of Br_2 are mixed in a 1.00 dm^3 vessel.

Calculate the concentration of HBr at equilibrium.

3. Hydrogen sulfide gas, H_2S , reacts to produce hydrogen gas and disulfur gas according to the following equation.



0.15 mol of H_2S is placed in a 1.00 dm^3 reaction vessel and allowed to reach equilibrium.

Determine the equilibrium concentration of $\text{S}_2(\text{g})$.

State the assumption made in the calculation.

1.

	$\text{I}_2(\text{g})$
Initial concentration (mol dm^{-3})	0.40
Change in concentration (mol dm^{-3})	-0.20
Equilibrium concentration (mol dm^{-3})	0.20

$$\begin{aligned}
 K_c &= \frac{[\text{IBr}]^2}{[\text{I}_2][\text{Br}_2]} \\
 &= \frac{[0.40]^2}{[0.20][0.20]} \\
 &= 4.0
 \end{aligned}$$

2.

	$\text{H}_2(\text{g})$
Initial concentration (mol dm^{-3})	0.600
Change in concentration (mol dm^{-3})	-x
Equilibrium concentration (mol dm^{-3})	$0.600 - x$

$$\begin{aligned}
 K_c &= \frac{[\text{HBr}]^2}{[\text{H}_2][\text{Br}_2]} \\
 &= \frac{[2x]^2}{[0.600 - x][0.600 - x]} \\
 \frac{(2x)^2}{(0.600 - x)^2} &= 11.2 \\
 \sqrt{\frac{(2x)^2}{(0.600 - x)^2}} &= \sqrt{11.2} \\
 \frac{(2x)}{(0.600 - x)} &= 3.346 \\
 x &= 0.376
 \end{aligned}$$

$$[\text{HBr}] = 2x = 0.752 \text{ mol dm}^{-3}$$

3. For this reaction the value of the equilibrium constant is very small, so the assumption is $[\text{reactant}]_{\text{initial}} \approx [\text{reactant}]_{\text{eqm}}$.

	$2\text{H}_2\text{S}(\text{g})$
Initial concentration (mol dm^{-3})	0.15
Change in concentration (mol dm^{-3})	$-2x$
Equilibrium concentration (mol dm^{-3})	≈ 0.15

$$\begin{aligned}
 K_c &= \frac{[\text{H}_2]^2[\text{S}_2]}{[\text{H}_2\text{S}]^2} \\
 9.30 \times 10^{-8} &= \frac{[2x]^2[x]}{[0.15]^2} \\
 4x^3 &= 2.09 \times 10^{-9} \\
 x &= 8.06 \times 10^{-4}
 \end{aligned}$$

$$[\text{S}_2] = 8.06 \times 10^{-4} \text{ mol dm}^{-3}$$